CYBERSECURITY LAB REPORT

Exercise 1: Port Scanning with Pentest Tools

Pentest-Tools.com • Port Scanner • Exercise 1

Target	scanme.nmap.org
Reported IP	45.33.32.156
Scan profile	Light • TCP • TCP SYN • Top 1,000 ports
Date	10 October 2026
Status	Completed — 2 open TCP ports reported

Scope note: This exercise uses the lab-provided `scanme.nmap.org` target. The lab material says this host permits port scanning for learning and testing, but not other penetration tests. Scan only systems for which you have authorization.

1. Objective

Perform a basic TCP port scan against the lab-provided target, identify open ports, and record the services and versions reported by the scanner.

2. Tools and Configuration

Parameter	Value
Platform / tool	Pentest-Tools.com / Port Scanner
Target / reported IP	scanme.nmap.org / 45.33.32.156
Scan type / protocol	Light / TCP
Port coverage	Top 1,000 ports
Scan technique	TCP SYN
Host-alive check	Enabled
Service/version detection	Enabled
OS detection / traceroute	Disabled / Disabled
Duration / status	13 seconds / Finished

The scanner notes that Light mode checks a limited range rather than all 65,535 ports. Results are not a complete inventory of every port on the host.

3. Results

The scan reported **2 open TCP ports** on the target.

Port	Protocol	State	Service	Product	Reported version / details
22	TCP	Open	SSH	OpenSSH	6.6.1p1 Ubuntu 2ubuntu2.13; Ubuntu Linux, protocol 2.0
80	TCP	Open	HTTP	Apache httpd	2.4.7 (Ubuntu)

Port 22 — SSH

SSH is commonly used for remote administration over an encrypted connection. The scanner identified OpenSSH and returned the version string shown in the table. A version string alone does not prove exploitability; exposure, configuration, patch status, and compensating controls would require an authorized assessment.

Port 80 — HTTP

HTTP is commonly used to serve web content. The scanner identified Apache httpd and reported version 2.4.7 (Ubuntu). An open HTTP port is not itself proof of a vulnerability.

4. Method

• Opened Pentest-Tools.com and selected New Scan.

• Selected Port Scanner and entered scanme.nmap.org.

• Confirmed Light scan, TCP, host-alive check enabled, custom scan time disabled, and workspace notifications selected.

• Confirmed authorization and started the scan.

• Reviewed the summary, results, and scan parameters, then recorded the findings and limitations.

5. Interpretation

The scan identified SSH on TCP/22 and HTTP on TCP/80. Service/version detection returned software identification details for both. The scan used TCP SYN and top-1,000-port coverage; OS detection and traceroute were disabled. These are point-in-time observations, and service fingerprints may not always be exact.

6. Limitations

• Light scan only; not a full 65,535-port scan.

• TCP was selected; UDP ports were not assessed.

• OS detection and traceroute were disabled.

• The scan does not establish whether either service is vulnerable.

• No exploitation, password testing, or other intrusive testing was performed.

7. Conclusion

Exercise 1 was completed successfully. The scan of scanme.nmap.org (45.33.32.156) reported two open TCP ports: 22/SSH and 80/HTTP. The exercise demonstrated basic scan configuration, service fingerprint interpretation, and technical evidence reporting.

8. Evidence to include in the repository

Keep the original scanner-export PDF and your own screenshots of the dashboard, scan setup, summary, and results. Redact any account details or secrets before publishing.

9. Skills demonstrated

Basic network reconnaissance • TCP port scanning concepts • Service/version identification • Interpreting scan parameters • Evidence collection • Responsible authorized testing

Portfolio summary: Conducted a light TCP SYN scan of the lab-authorized scanme.nmap.org target using Pentest-Tools.com. Identified two open services (SSH on TCP/22 and HTTP on TCP/80), reviewed service/version fingerprints, documented scan configuration and limitations, and preserved the output as evidence. This exercise focused on reconnaissance and reporting, not exploitation.